From Gravitational Collapse to Pseudoscalaron Inflation
Lecture Notes

Physics Research Preparation

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1 Lecture 1: FLRW Geometry and Gravitational Collapse

1.1 Physical picture

We study an idealised, homogeneous, isotropic, spherical cloud of pressureless matter. Fluid elements keep fixed comoving coordinates while the physical distances between them can change. It is not the coordinate labels that shrink; the spatial metric evolves. We work in units $c = 1$, use signature $(-,+,+,+)$ and take t to be the proper time of comoving fluid observers. A homogeneous interior is not itself a global black-hole spacetime.

1.2 FLRW metric

Starting from $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$, homogeneity and isotropy motivate the metric

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)\right]. \tag{1}$$

Here $a(t)$ is the scale factor and k is the spatial-curvature parameter. The metric is a symmetry-based ansatz; its evolution must be determined from field equations.

1.3 Areal radius of the cloud

For the boundary at fixed comoving coordinate r_b, the angular metric is $ds^2_{\text{angular}} = a^2 r_b^2 d\Omega^2$. Comparing with $R^2 d\Omega^2$ gives

$$\boxed{R(t) = a(t) r_b}, \qquad \boxed{\dot{R}(t) = r_b \dot{a}(t)}. \tag{2}$$

Thus r_b remains fixed, but the physical areal radius R decreases if $\dot{a} < 0$. Note: the areal radius is not generally equal to radial proper distance from the centre when $k \neq 0$.

1.4 Gravitational dynamics

For a perfect fluid with energy density ρ and pressure p, and with $\Lambda = 0$, the FLRW reduction of Einstein's equations gives

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho, \qquad H \equiv \frac{\dot{a}}{a}, \tag{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p), \tag{4}$$

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{5}$$

Contraction requires $H < 0$. A negative acceleration $\ddot{a} < 0$ is not sufficient: an expanding model can also have $\ddot{a} < 0$.

1.5 Dust density scaling: step-by-step

For pressureless dust, $p = 0$, so

$$\dot{\rho} + 3\frac{\dot{a}}{a}\rho = 0, \tag{6}$$

$$\frac{d\rho}{\rho} = -3\frac{da}{a}, \tag{7}$$

$$\ln(\rho/\rho_0) = -3\ln(a/a_0), \tag{8}$$

$$\boxed{\rho(a)} = \boxed{\rho_0 (a_0/a)^3}. \tag{9}$$

A halving of a increases the density by a factor of eight. For an initially static dust cloud with $\rho_i > 0$, the first Friedmann equation requires $k > 0$ (with $\Lambda = 0$).

1.6 Vocabulary

Metric tensor
$g_{\mu\nu}$ defines spacetime intervals.

Scale factor
$a(t)$ changes physical separations between comoving points.

Comoving coordinate
Fixed coordinate label attached to a fluid element.

Proper time
Time recorded by a comoving clock.

Areal radius
$R = \sqrt{A/(4\pi)}$ for a spherical surface.

Hubble parameter
$H = \dot{a}/a$, negative during contraction.

1.7 Literature and scope

Carroll explains FLRW and Friedmann equations in general relativity. Oppenheimer and Snyder (1939) analyse classical homogeneous dust collapse. Hashemi, Jalalzadeh and Ziaie investigate a related homogeneous spin-fluid collapse model in Einstein–Cartan gravity. The torsion-induced bounce is *not* derived in this first lecture.

1.8 Lecture 1 recap

1. FLRW is a homogeneous/isotropic interior metric ansatz.
2. The boundary's physical areal radius is $R = ar_b$.
3. Comoving coordinate labels stay fixed; metric distances evolve.
4. Collapse requires $H < 0$; $\ddot{a} < 0$ alone does not establish contraction.
5. For dust, $\rho \propto a^{-3}$; the density rises as the cloud contracts.
6. Establishing a complete black-hole spacetime requires additional global matching.

1.9 One-minute oral explanation

> The starting point is an idealised homogeneous spherical collapse model, similar to Oppenheimer–Snyder collapse. We describe the interior with an FLRW metric. Its scale factor determines physical separations between comoving matter elements. During contraction the scale factor decreases, so the Hubble parameter is negative. For pressureless matter, energy conservation implies $\rho \propto a^{-3}$. We then want to investigate how Einstein–Cartan spin–torsion effects modify this classical picture.

1.10 Sources

- S. M. Carroll, *Lecture Notes on General Relativity*, https://arxiv.org/abs/gr-qc/9712019.
- J. R. Oppenheimer and H. Snyder, *On Continued Gravitational Contraction* (1939).
- Hashemi, Jalalzadeh and Ziaie, *Collapse and dispersal of a homogeneous spin fluid in Einstein–Cartan theory*, https://arxiv.org/abs/1407.4103.

2 Lecture 2: Einstein–Cartan Gravity, Spin and Torsion

2.1 Physical motivation

General relativity uses the torsion-free Levi–Civita connection. Einstein–Cartan (EC) gravity uses a metric-compatible connection which can also have torsion. In EC theory, intrinsic angular momentum (spin) sources torsion through an *algebraic* field equation. In minimal EC gravity, torsion is not an independently propagating wave-like field. We use $c = 1$, $\kappa = 8\pi G$ and metric signature $(-,+,+,+)$.

2.2 Connection and torsion

For a vector field V^ρ,

$$\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma^\rho{}_{\mu\nu} V^\nu. \tag{10}$$

The torsion tensor is the antisymmetric part of the connection:

$$\boxed{T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu} = 2\Gamma^\rho{}_{[\mu\nu]}}. \tag{11}$$

In GR the Levi–Civita connection is symmetric:

$$\{^\rho{}_{\mu\nu}\} = \frac{1}{2} g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}), \qquad \{^\rho{}_{\mu\nu}\} = \{^\rho{}_{\nu\mu}\}\,. \tag{12}$$

For a metric-compatible torsionful connection we define the contortion by

$$\Gamma^\rho{}_{\mu\nu} = \{^\rho{}_{\mu\nu}\} + K^\rho{}_{\mu\nu}. \tag{13}$$

Trautman and Bréchet *use different sign and factor conventions* for the contortion and torsion. Do not combine their component equations without translating conventions.

2.3 Why intrinsic spin matters

An energy–momentum tensor describes translational momentum and energy. Matter can additionally carry intrinsic angular momentum (quantum spin), which is distinct from bulk rotation or fluid vorticity. In the EC variational principle, the tetrad (or metric variables) and spin connection are varied independently, producing a gravitational equation and a Cartan spin–torsion equation.

A schematic minimal EC action is

$$S[e,\omega,\psi] = \int d^4x\, e \left[\frac{1}{2\kappa}\widetilde{R}(e,\omega) + \mathcal{L}_m(e,\omega,\psi)\right], \tag{14}$$

where $e = \det(e_\mu{}^a)$ and $\widetilde{R}$ is the scalar curvature of the torsionful connection. Bréchet et al. write the analogous action with a cosmological constant in their Eq. (11).

2.4 Variation: the field equations

Variation with respect to the gravitational coframe gives an Einstein-like equation written using the full connection; variation with respect to the spin connection gives an algebraic equation of the form

$$\boxed{T^\rho{}_{\mu\nu} + \text{torsion-trace terms} = \kappa\, \tau^\rho{}_{\mu\nu}.} \tag{15}$$

Here $\tau^\rho{}_{\mu\nu}$ is the intrinsic spin current. The detailed index positions and trace signs depend on convention; compare Trautman Eqs. (23)–(25) and Bréchet et al. Eqs. (12)–(15). The crucial point is the absence of derivatives of torsion in this *minimal* Cartan equation:

$$\boxed{\text{spin current } \longrightarrow \text{ algebraically determined torsion.}} \tag{16}$$

If the spin current vanishes, torsion vanishes within minimal EC theory, and the vacuum equations reduce to those of GR.

2.5 Weyssenhoff spin fluid

For a macroscopic spinning fluid, Bréchet et al. introduce the antisymmetric spin density $S_{\mu\nu} = -S_{\nu\mu}$ and the current

$$\tau^{\lambda}{}_{\mu\nu} = u^{\lambda} S_{\mu\nu}, \tag{17}$$

where u^{λ} is the fluid four-velocity. Their Frenkel condition is

$$\boxed{S_{\mu\nu} u^{\nu} = 0}. \tag{18}$$

It states that the spin tensor is spatial in the fluid's rest frame. Under this condition the torsion trace vanishes in their model, and their Cartan equation reduces to an algebraic relation between torsion and $u^{\lambda} S_{\mu\nu}$ (their Eq. (21)). A scalar spin measure is

$$\boxed{S^2 = \frac{1}{2} S_{\mu\nu} S^{\mu\nu}}. \tag{19}$$

2.6 Eliminating torsion: effective energy and pressure

Because torsion is algebraically related to spin, it can be eliminated from the minimal EC field equations. The resulting Levi–Civita Einstein equations contain *additional terms quadratic in spin*. This is discussed by Trautman in Eqs. (30)–(32).

In the particular Weyssenhoff-fluid conventions of Bréchet et al. (and with cosmological constant set to zero), Eq. (25) reads

$$\boxed{\rho_s = \rho - \kappa S^2}, \qquad \boxed{p_s = p - \kappa S^2}. \tag{20}$$

These expressions are *model- and convention-dependent*; the full effective stress tensor also contains spin-gradient terms (their Eq. (24)). They are not a universal replacement for every Einstein–Cartan matter source or every collapse geometry.

2.7 Connection to gravitational collapse

For conserved particle number in a homogeneous comoving fluid, $n \propto a^{-3}$. Under additional assumptions about the spin state and averaging, the mean squared spin density scales as $S^2 \propto n^2 \propto a^{-6}$. Thus the spin-squared contribution may grow faster under contraction than the ordinary dust energy density $\rho \propto a^{-3}$.

Important limitation: these scalings do not alone demonstrate a nonsingular bounce. One must insert a consistent effective stress tensor into the gravitational equations and verify both $H(t_B) = 0$ and $\dot{H}(t_B) > 0$. That is the task of Lecture 3.

2.8 Mathematical clarification: no circular definition

The relations

$$T^{\rho}{}_{\mu\nu} = 2\Gamma^{\rho}{}_{[\mu\nu]}, \qquad \Gamma^{\rho}{}_{\mu\nu} = \{^{\rho}{}_{\mu\nu}\} + K^{\rho}{}_{\mu\nu} \tag{21}$$

are *definitions and a change of geometric variables*, not simultaneous independent dynamical equations. One can start with a metric g and a metric-compatible connection Γ and calculate torsion, or use the equivalent data (g, T) to reconstruct Γ. Metric compatibility fixes the required form of contortion. With $\nabla_{\mu} V^{\rho} = \partial_{\mu} V^{\rho} + \Gamma^{\rho}{}_{\mu\nu} V^{\nu}$ and $K = \Gamma - \{\,\}$, lowering the first index gives

$$T_{\rho\mu\nu} = K_{\rho\mu\nu} - K_{\rho\nu\mu}, \qquad K_{\rho\mu\nu} = -K_{\nu\mu\rho}, \tag{22}$$

$$\boxed{K_{\rho\mu\nu}} = \boxed{\frac{1}{2}\left(T_{\rho\mu\nu} - T_{\mu\nu\rho} + T_{\nu\rho\mu}\right)}. \tag{23}$$

The torsion equation $T = 2\Gamma_{[\mu\nu]}$ is a definition. The algebraic relation $K = K(T)$ follows from this definition and metric compatibility. Only the *Cartan field equation*, derived from the action, determines torsion from the matter spin current in minimal EC gravity. One cannot simply use $\Gamma = \{\} + T$ because that would in general double the antisymmetric connection difference and fail to satisfy the required index symmetries.

2.9 Mathematical clarification: the Frenkel condition

For a Weyssenhoff spin fluid, $S_{\mu\nu} = -S_{\nu\mu}$ and $S_{\mu\nu}u^\nu = 0$. In the local fluid rest frame $u^\nu = (1, 0, 0, 0)$, so

$$0 = S_{\mu\nu}u^\nu = S_{\mu 0}. \tag{24}$$

Antisymmetry then implies $S_{0i} = S_{i0} = 0$, but the spatial components S_{12}, S_{13}, S_{23} may be nonzero. These are three independent spin components. The condition is part of the Weyssenhoff-fluid model and should not be confused with requiring aligned spins or vanishing spin density. When averaging randomly oriented spins, $\langle S_{\mu\nu}\rangle = 0$ does not necessarily imply $\langle S_{\mu\nu}S^{\mu\nu}\rangle = 0$.

2.10 Technical vocabulary

Affine connection
$\Gamma^\rho{}_{\mu\nu}$ defines covariant derivatives and parallel transport.

Levi–Civita connection
Torsion-free connection uniquely determined by the metric.

Torsion tensor
$T^\rho{}_{\mu\nu} = 2\Gamma^\rho{}_{[\mu\nu]}$.

Contortion
The difference of the metric-compatible full connection and Levi–Civita connection, with sign conventions stated.

Intrinsic spin
Internal angular momentum of matter, not the same as macroscopic rotation.

Spin current
Source of torsion in the Cartan field equation.

Weyssenhoff fluid
Phenomenological continuum model incorporating an antisymmetric spin density.

Frenkel condition
$S_{\mu\nu}u^\nu = 0$.

2.11 Lecture 2 recap

1. GR uses a torsion-free Levi–Civita connection; minimal EC permits torsion alongside curvature.
2. Torsion is the antisymmetric part of the connection and is sourced by intrinsic spin.
3. In *minimal EC*, the Cartan equation is algebraic: torsion does not independently propagate.
4. A Weyssenhoff fluid models spinning matter with $S_{\mu\nu}$ and the Frenkel condition.
5. Eliminating torsion produces effective spin-squared corrections to the GR matter equations.

6. Growing spin-squared terms during collapse motivate, but do not by themselves prove, a bounce.

2.12 One-minute oral explanation

> Einstein–Cartan gravity extends general relativity by allowing spacetime torsion in addition to curvature. In minimal Einstein–Cartan theory, intrinsic spin sources torsion through an algebraic field equation. This is different from a new propagating field. For a Weyssenhoff spin fluid, we can eliminate torsion and obtain effective corrections quadratic in spin. These corrections may become important during gravitational collapse, because the squared spin density can grow faster than the ordinary matter density. The next question is whether they are sufficient to produce a consistent bounce.

2.13 Sources and research boundary

- A. Trautman, *Einstein–Cartan Theory*, https://arxiv.org/abs/gr-qc/0606062, especially pp. 3–6, Eqs. (23)–(25) and (30)–(32).
- S. D. Bréchet et al., *Weyssenhoff fluid dynamics in general relativity using a 1+3 covariant approach*, https://arxiv.org/abs/0706.2367, especially Eqs. (11)–(25).

A *nonlinear Holst extension* can change the torsion sector's degrees of freedom and is *not* derived by these minimal EC equations; it is reserved for a later lecture.

3 Lecture 3: Spin–Torsion Bounce

To be developed after the lesson.

4 Lecture 4: Nonlinear Holst Theory

To be developed after the lesson.

5 Lecture 5: Pseudoscalaron Inflation

To be developed after the lesson.